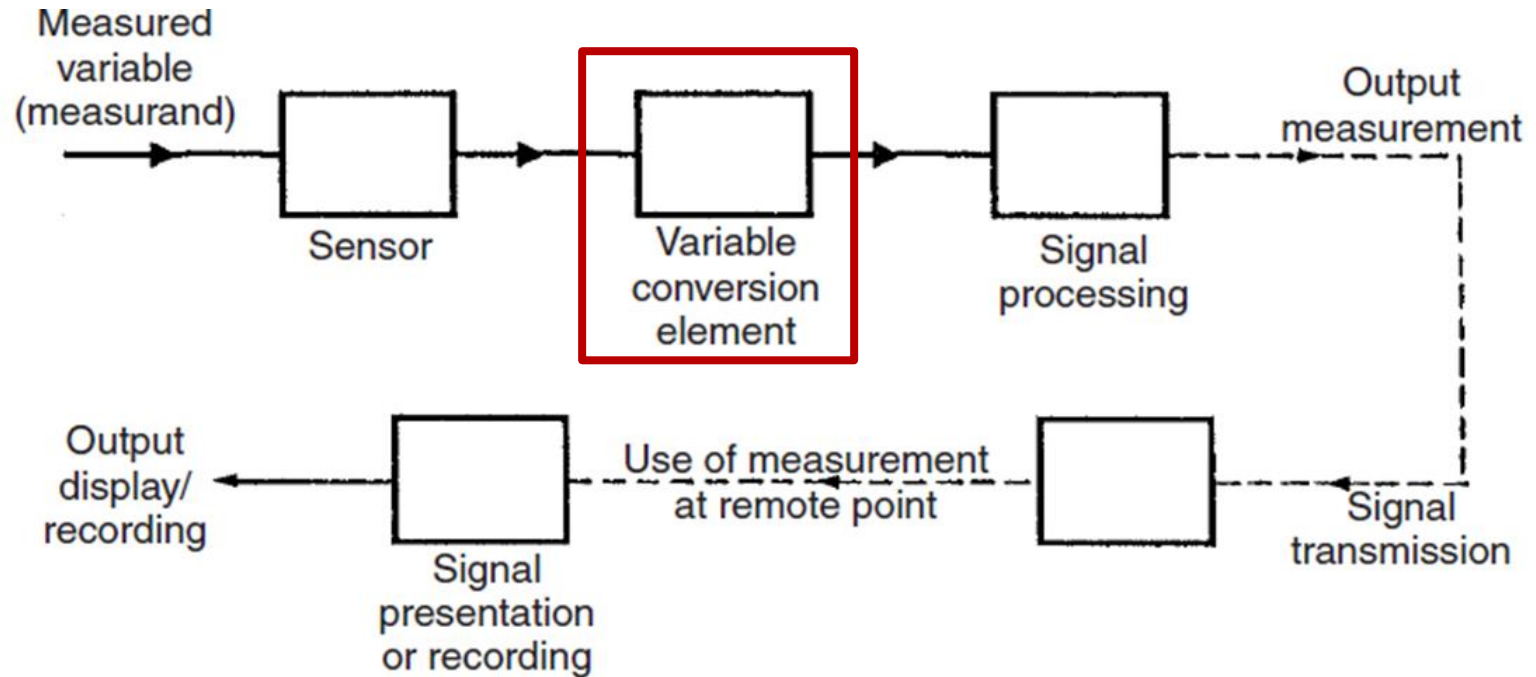


Variable Conversion Elements

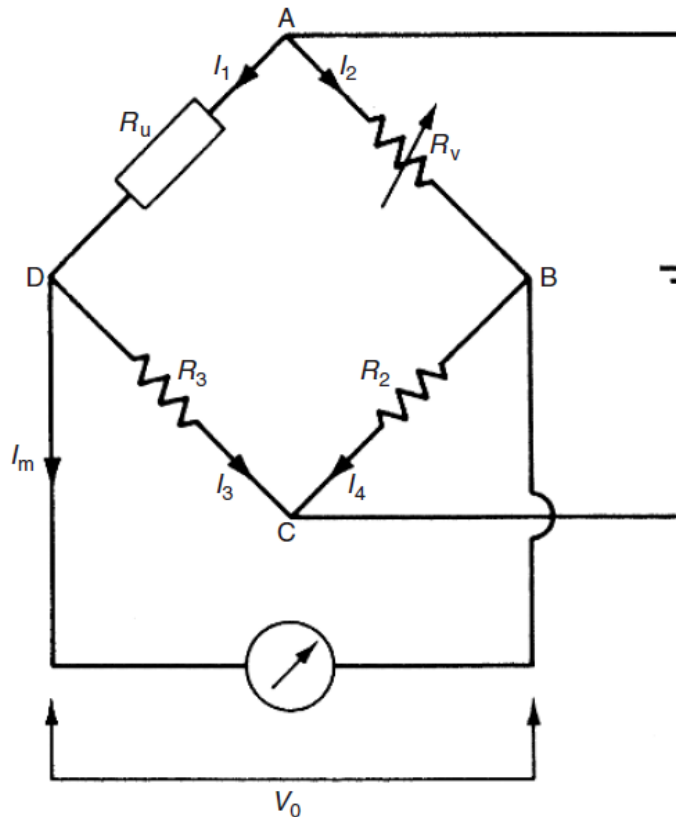


Variable Conversion Elements



Bridge circuits

□ Balanced, DC bridge (Wheatstone bridge)



- Unknown resistance R_u
- Two equal value resistor R_2 and R_3
- Variable resistor R_v

Assume $I_m = 0$

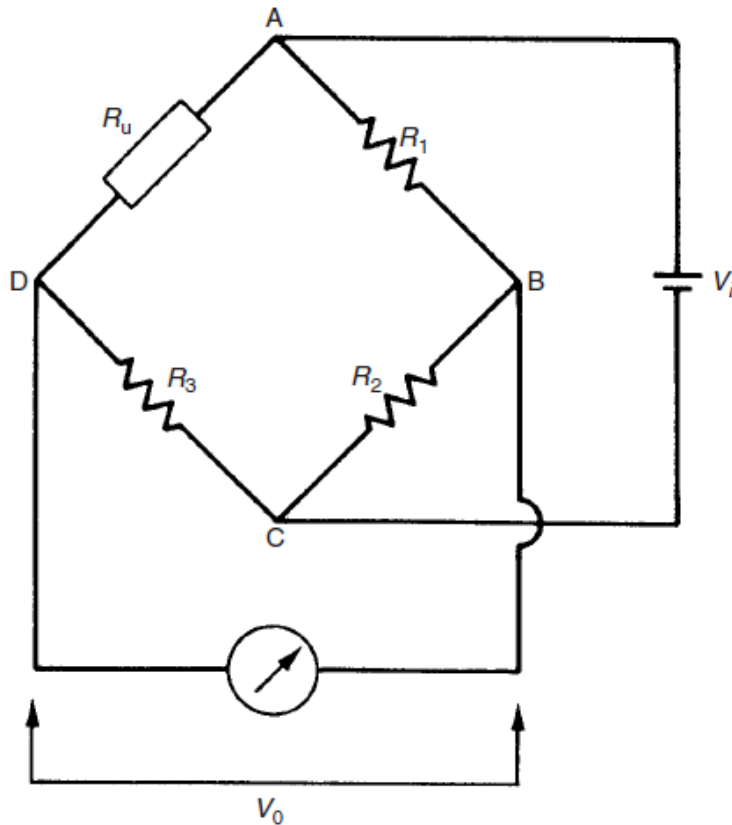
$$I_1 = I_3 \text{ and } I_2 = I_4$$

$$R_u R_2 = R_3 R_v \text{ or } R_u = \frac{R_3 R_v}{R_2}$$

Fig. 7.1 Analysis of Wheatstone bridge.

Bridge circuits

□ Unbalanced DC bridge



- Unknown resistance R_u
- Two equal value resistor R_2 and R_3
- R_1 has the same value of the nominal value of R_u

$$V_0 = V_i \left(\frac{R_u}{R_u + R_3} - \frac{R_1}{R_1 + R_2} \right)$$

V_0 varies non-linearly with R_u

Fig. 7.2 Deflection-type d.c. bridge.

Bridge circuits

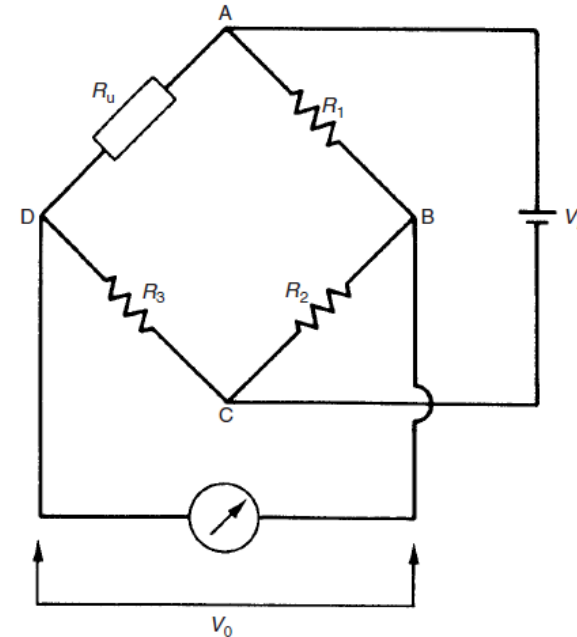
□ Deflection-type DC bridge

Example

Một loại đầu dò áp suất được thiết kế để đo áp suất trong phạm vi 0-10bar. Cảm biến áp suất có điện trở danh định là 120Ω và tạo thành một nhánh của mạch cầu Wheatstone, với ba nhánh còn lại, mỗi nhánh có điện trở 120Ω . Đầu ra của mạch cầu được đo bằng một thiết bị có trở kháng đầu vào vô hạn.

Để hạn chế tác động do nhiệt, dòng điện cực đại cho phép qua cảm biến áp suất là 30mA, hãy tính điện áp kích mạch cầu lớn nhất.

Nếu độ nhạy của máy đo biến dạng là $338m\Omega/\text{bar}$ và sử dụng điện áp kích thích mạch cầu lớn nhất, hãy tính điện áp đầu ra của mạch cầu khi đo áp suất 10 bar.



Bridge circuits

□ Deflection-type DC bridge

Example

$$R_1 = R_2 = R_3 = 120\Omega$$

Defining I_1 to be the current flowing in path ADC of the bridge

$$V_i = I_1(R_u + R_3)$$

At balance, $R_u = 120$, the maximum value allowable for I_1 is 0.03 A.

$$V_i = 0.03(120 + 120) = 7.2 \text{ V}$$

For a pressure of 10 bar applied, the resistance change is 3.38 Ω , i.e. R_u is then equal to 123.38 Ω .

$$V_0 = V_i \left(\frac{R_u}{R_u + R_3} - \frac{R_1}{R_1 + R_2} \right) = 7.2 \left(\frac{123.38}{243.38} - \frac{120}{240} \right) = 50 \text{ mV}$$

Bridge circuits

□ AC balanced impedance bridges

$$I_1 R_1 = I_2 R_2; I_1 Z_u = I_2 Z_v$$
$$Z_u = \frac{Z_v R_1}{R_2}$$

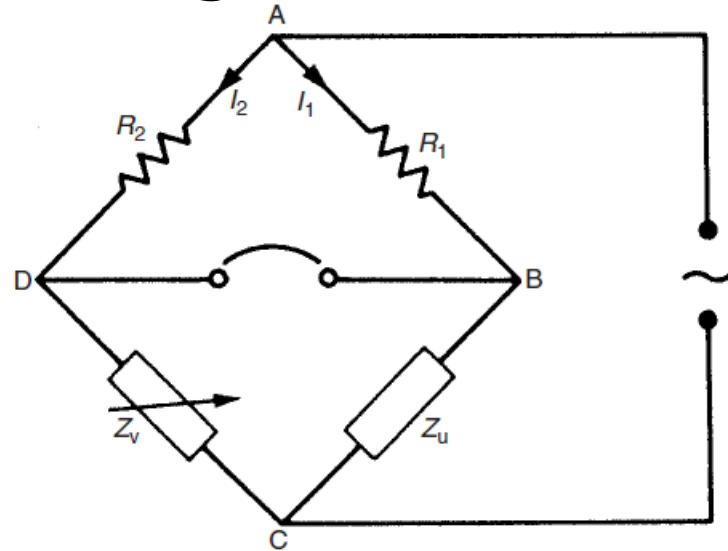
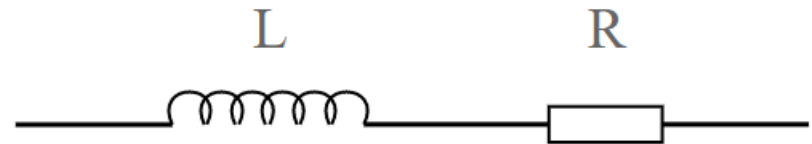


Fig. 7.7 Null-type impedance bridge.

- If Z_u is capacitive, $Z_u = 1/j\omega C_u$, variable capacitance.
- If Z_u is inductive, $Z_u = R_u + j\omega L_u$, variable-resistance and a variable-inductance.

Bridge circuits

□ AC null-type impedance bridges



Q factor = inductance/resistance.

$$Q = \frac{\omega L}{R}$$

Bridge circuits

□ Maxwell bridge

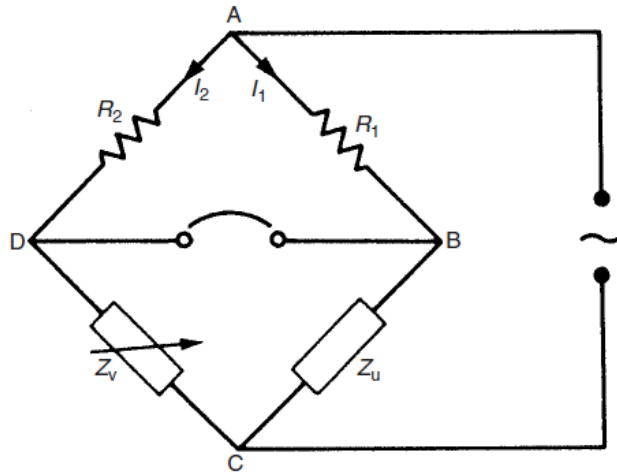


Fig. 7.7 Null-type impedance bridge.

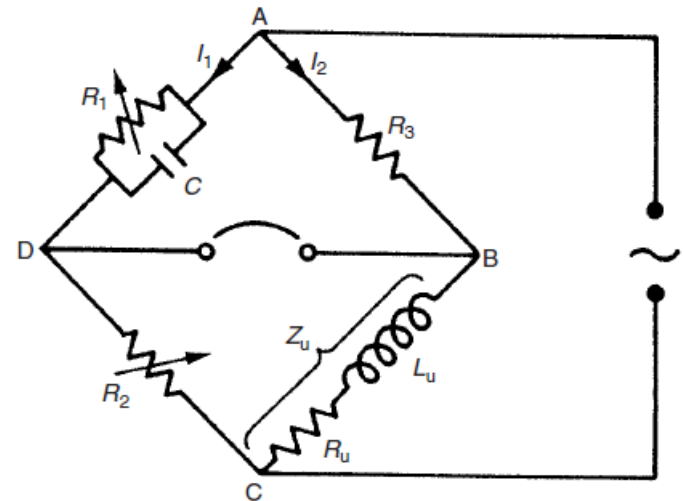


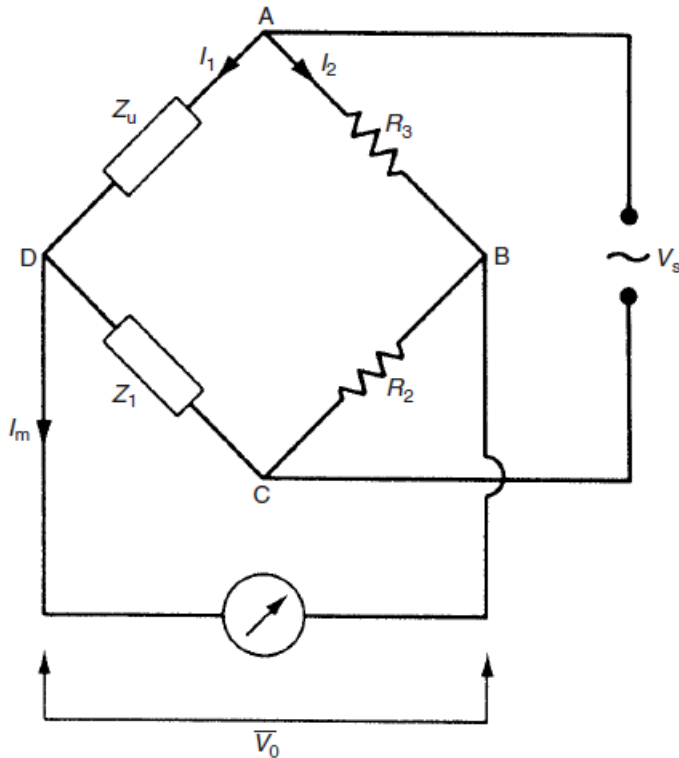
Fig. 7.8 Maxwell bridge.

$$R_u = \frac{R_2 R_3}{R_1}; L_u = R_2 R_3 C$$

$$Q = \frac{\omega L_u}{R_u} = \omega C R_1 \rightarrow \text{measure the } Q \text{ value of a coil.}$$

Bridge circuits

□ Unbalanced AC bridge



For capacitance measurement:

$$Z_u = \frac{1}{j\omega C}; \quad Z_1 = \frac{1}{j\omega C}$$

For inductance measurement:

$$Z_u = j\omega L_u; \quad Z_1 = j\omega L$$
$$V_0 = V_s \left(\frac{Z_u}{Z_1 + Z_u} - \frac{R_3}{R_2 + R_3} \right)$$

For capacitances: $V_0 = V_s \left(\frac{C_1}{C_1 + C_u} - \frac{R_3}{R_2 + R_3} \right)$

For inductances: $V_0 = V_s \left(\frac{L_u}{L_1 + L_u} - \frac{R_3}{R_2 + R_3} \right)$

Inductance measurement

- Output in the form of a change in inductance: inductive displacement sensor
- Measured by an AC bridge circuit

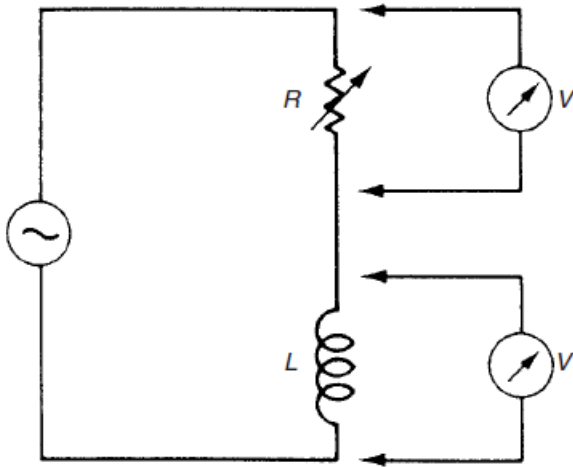


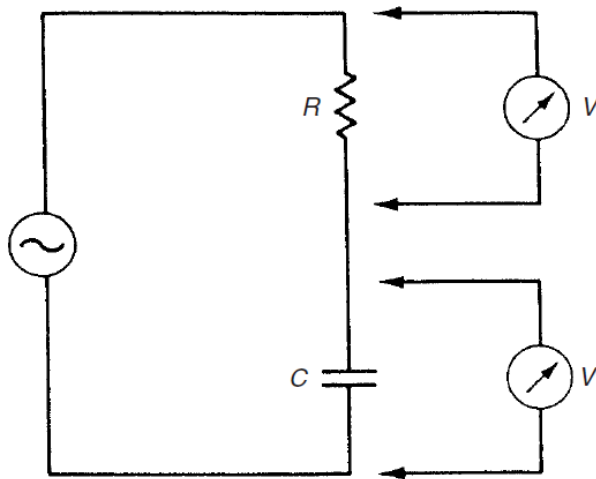
Fig. 7.13 Approximate method of measuring inductance.

- Connecting the unknown inductance in series with a variable resistance in a circuit excited with a sinusoidal voltage.
- The variable resistance is adjusted until the voltage measured across the resistance is equal to that measured across the inductance.

$$L = \frac{\sqrt{R^2 - r^2}}{2\pi f}$$

Capacitance measurement

- Output in the form of a change in capacitance: capacitive displacement sensor, capacitive moisture meter and capacitive hygrometer.
- Measured by an AC bridge circuit.

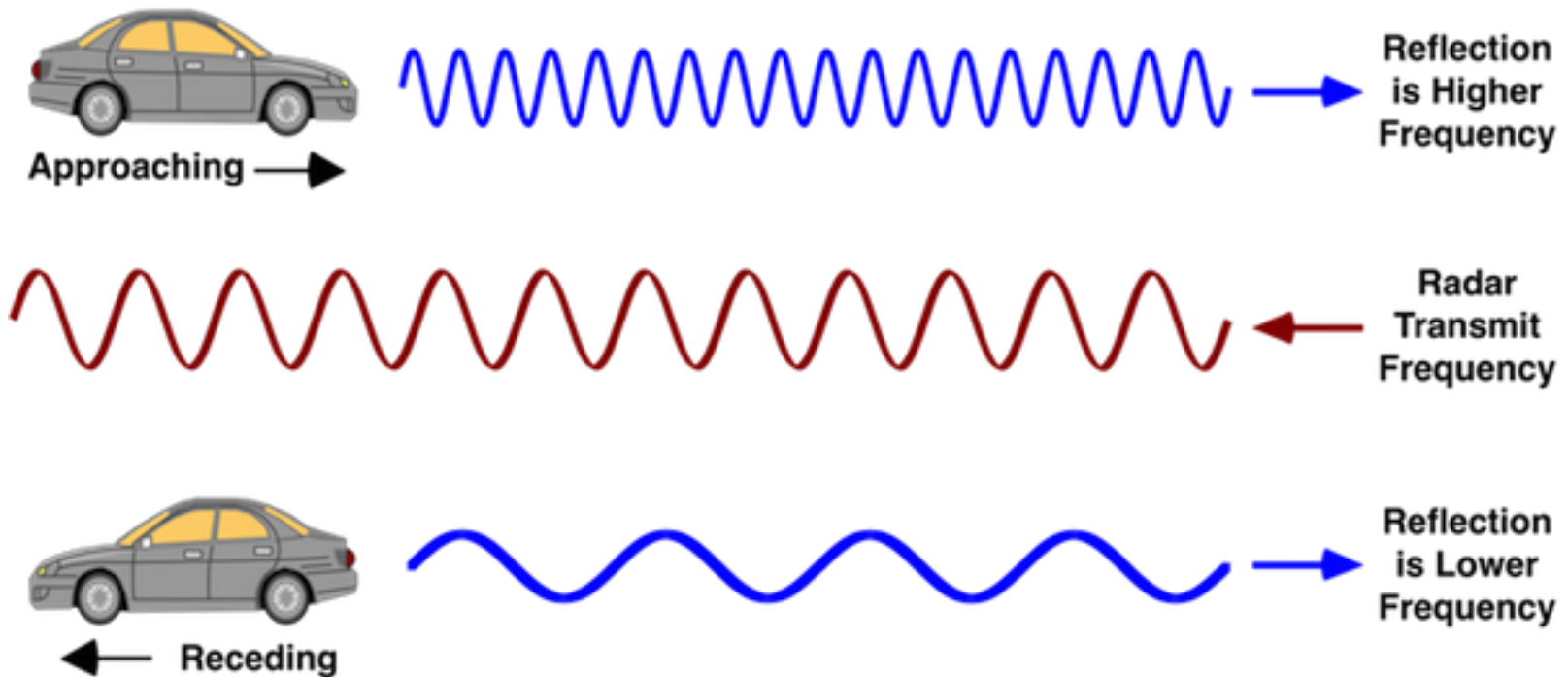


- Connecting the unknown capacitor in series with a variable resistance in a circuit excited with a sinusoidal voltage.
- An AC voltmeter is used to measure the voltage drop across both the resistor and the capacitor

$$C = \frac{V_r}{2\pi f R V_c}$$

Fig. 7.14 Approximate method of measuring capacitance.

Frequency measurement



Doppler effect

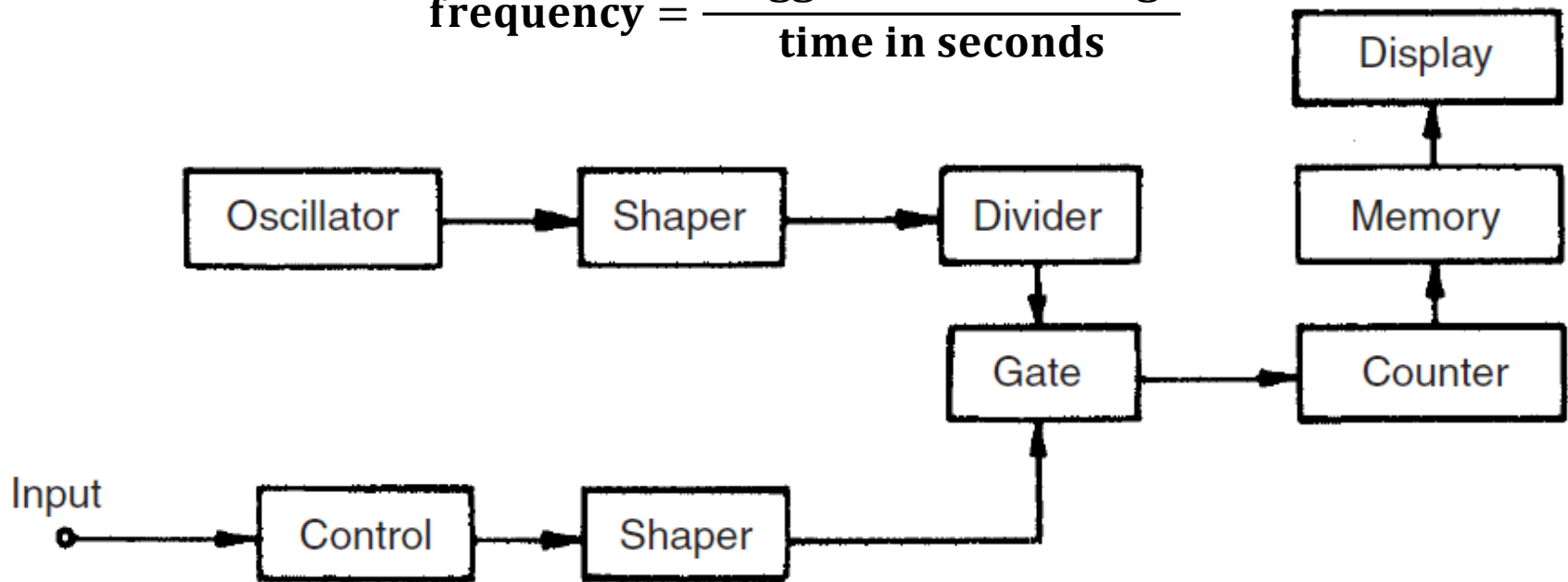
$$f_o = \left(\frac{c + v_o}{c + v_s} \right) f_s$$

- f_o : observer frequency of sound
- c : speed of sound waves
- v_o : observer velocity
- v_s : source velocity
- f_s : actual frequency of sound waves

Frequency measurement

□ Digital counter-timers

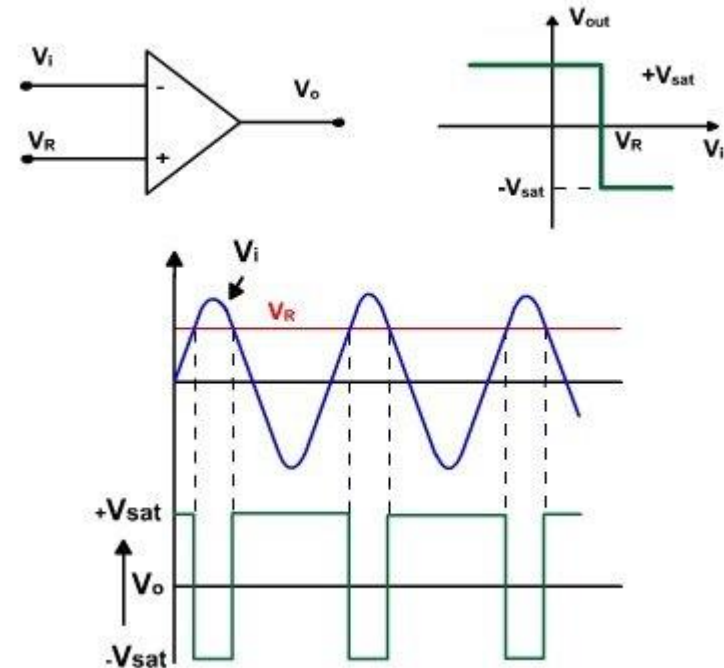
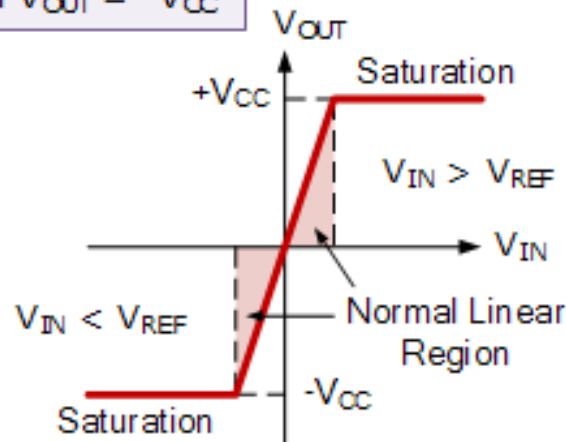
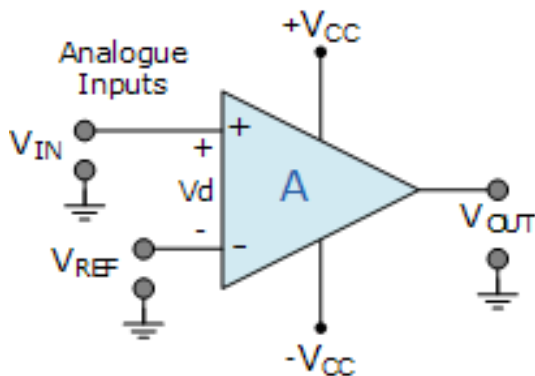
$$\text{frequency} = \frac{\text{Trigger level crossings}}{\text{time in seconds}}$$



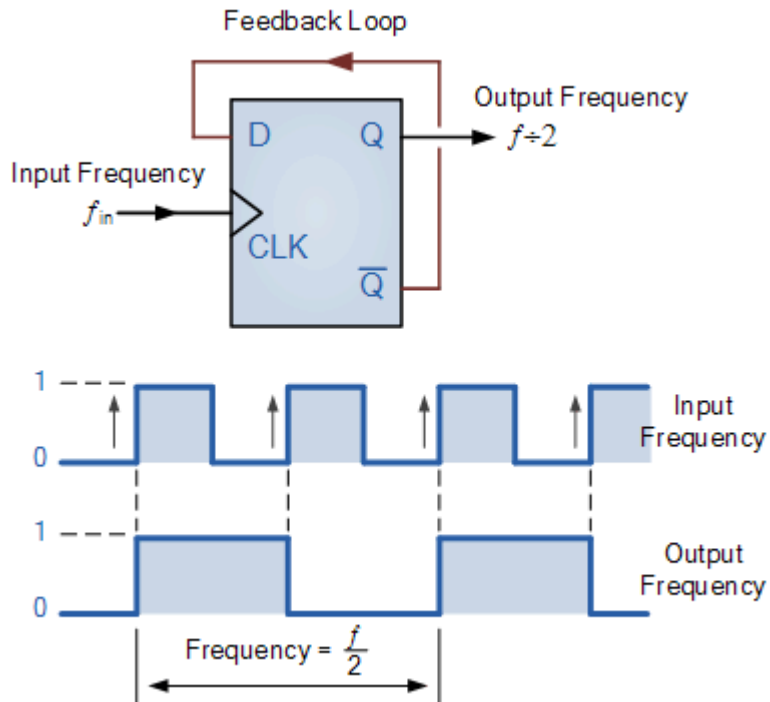
The accuracy of measurement depends upon how far the unknown frequency is above the reference frequency.

Frequency measurement

If $V_{IN} > V_{REF}$ then $V_{OUT} = +V_{CC}$
If $V_{IN} < V_{REF}$ then $V_{OUT} = -V_{CC}$

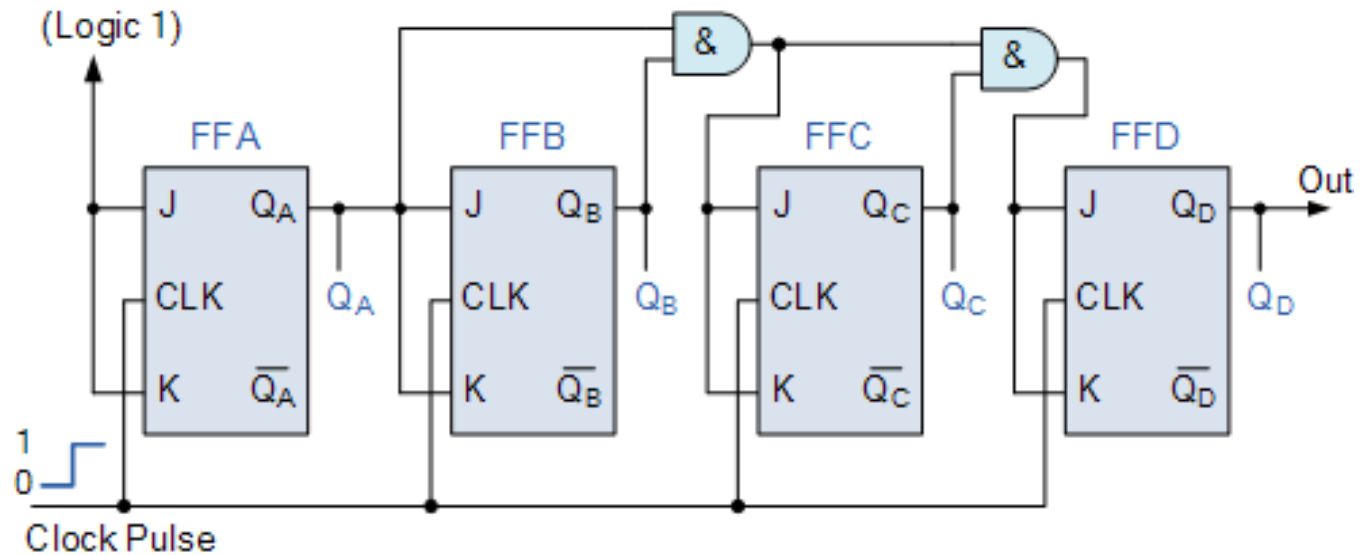


Frequency measurement



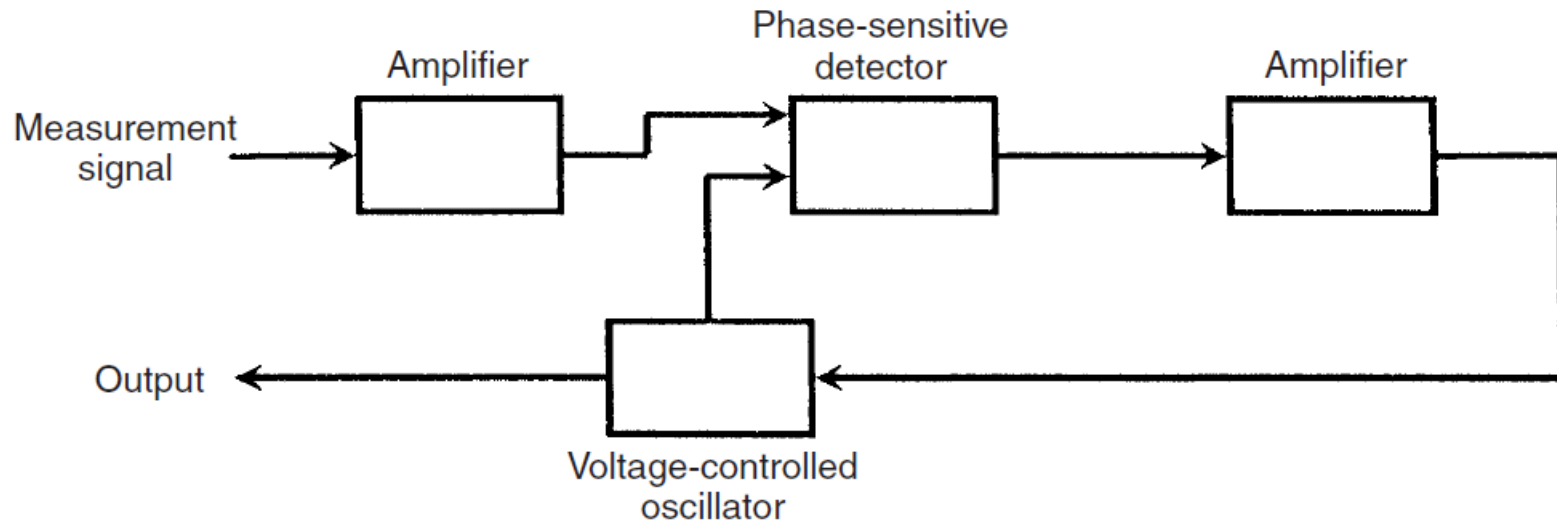
Clk	D	Q		Description
$\downarrow \gg 0$	X	Q	$\overline{Q}$	Memory no change
$\uparrow \gg 1$	0	0	1	Reset Q $\gg 0$
$\uparrow \gg 1$	1	1	0	Set Q $\gg 1$

Frequency measurement



Frequency measurement

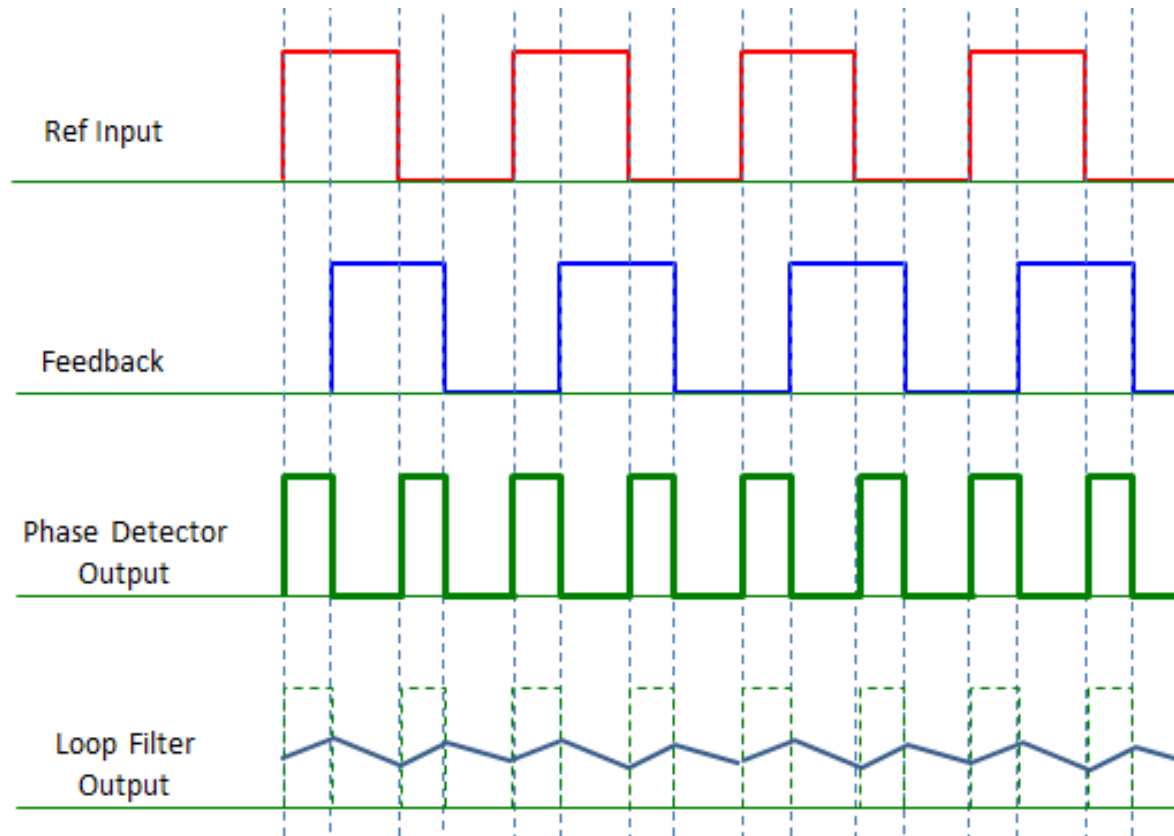
□ Phase-locked loop



The DC output from the VCO is then proportional to the input signal frequency

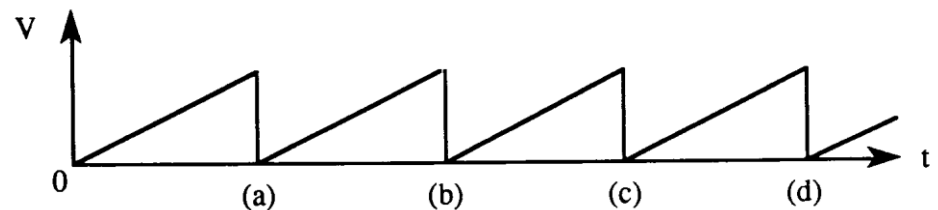
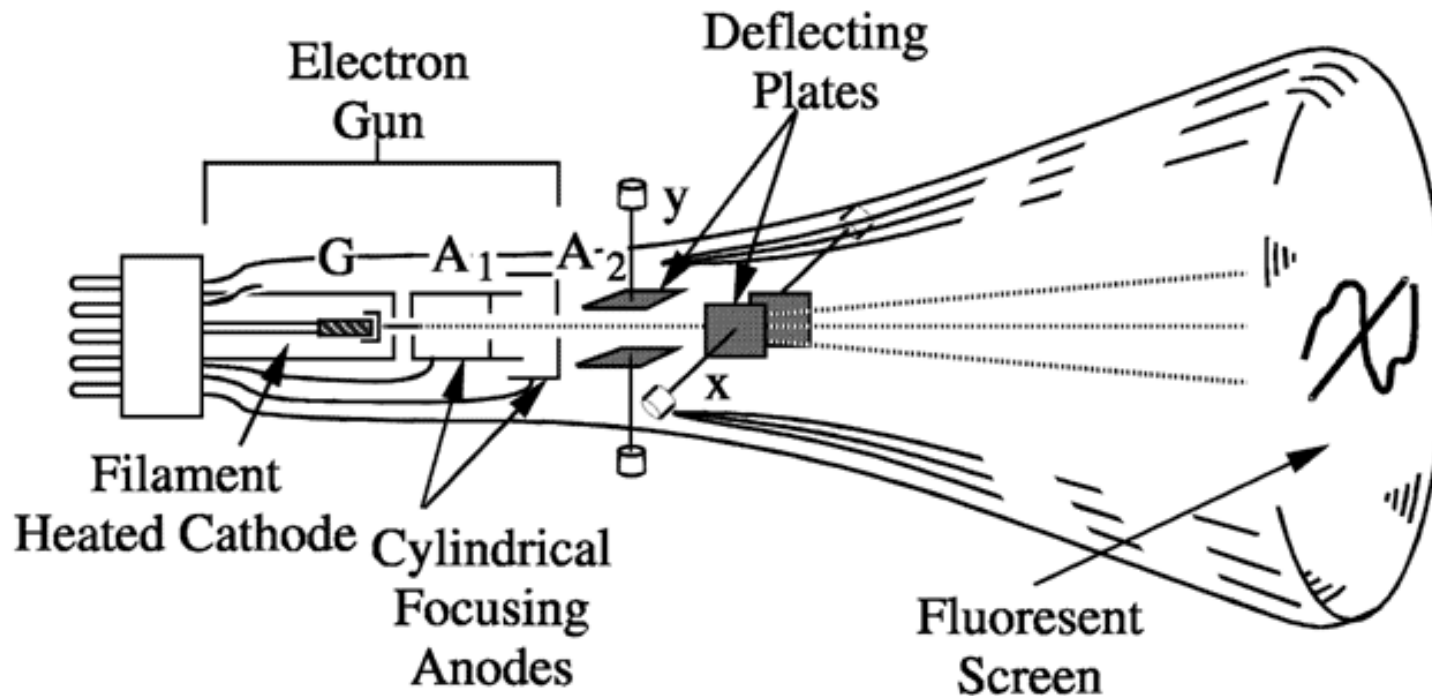
Frequency measurement

□ Phase-locked loop



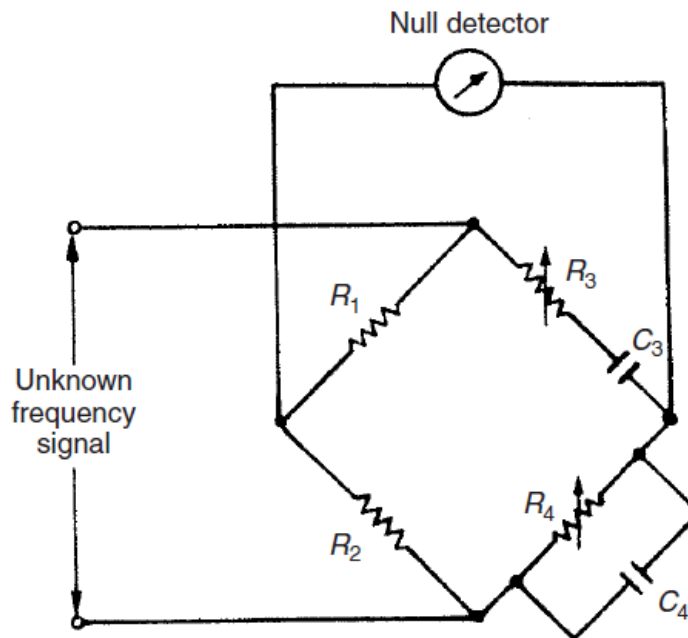
Frequency measurement

□ Cathode ray oscilloscope



Frequency measurement

□ The Wien bridge



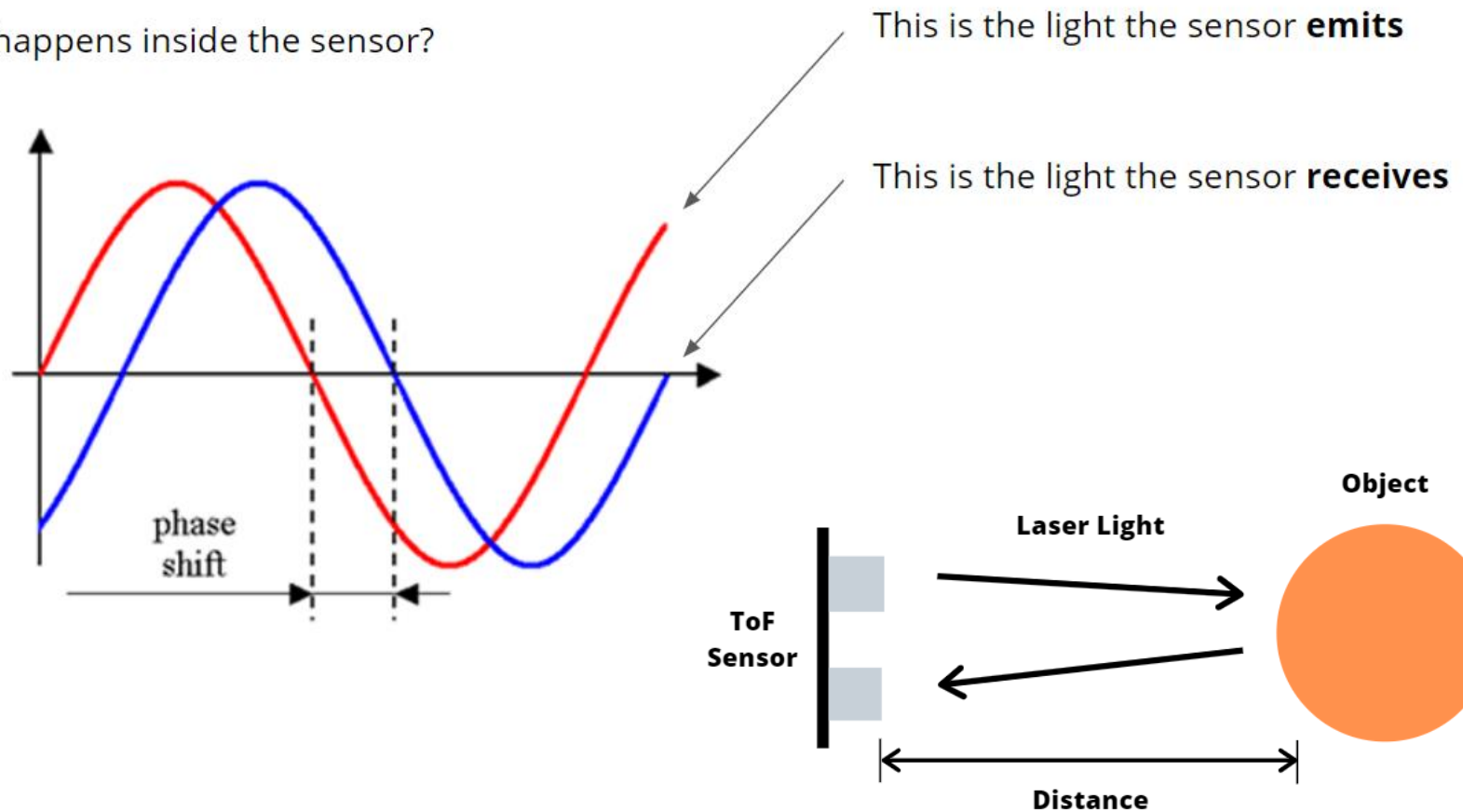
$$f = \frac{1}{2\pi R_3 C_3}$$

Fig. 7.19 Wien bridge.

The instrument is very accurate at audio frequencies, but at higher frequencies errors due to losses in the capacitors become significant.

Phase measurement

What happens inside the sensor?



Phase measurement

❑ Electronic counter-timer

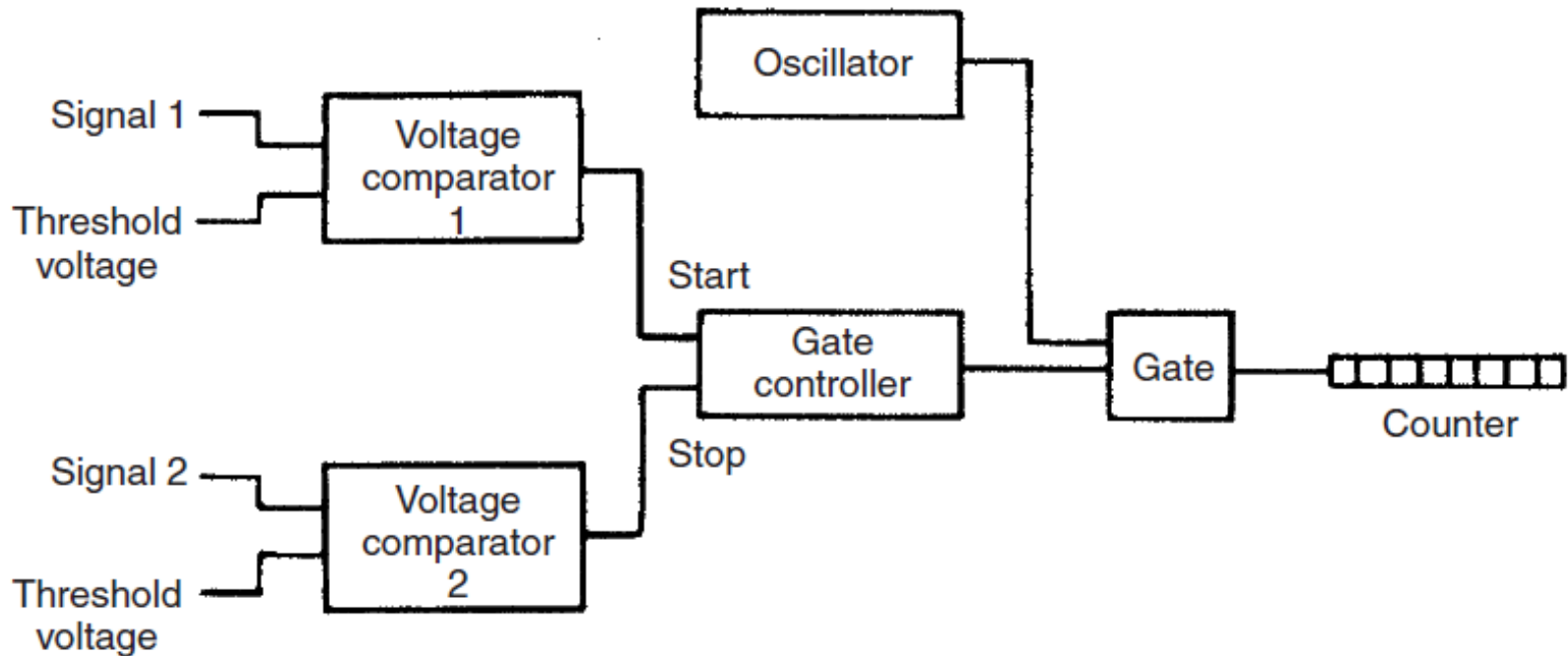


Fig. 7.20 Phase measurement with digital counter-timer.

Voltage-Controlled Oscillator (VCO)

